

React JS Notes (Complete Guide)

1. Introduction: React is a JavaScript library used to build user interfaces using components.

2. Key Concepts:

- Components: Reusable building blocks
- JSX: JavaScript XML syntax
- Virtual DOM: Efficient UI updates

3. Project Setup:

- CRA: `npx create-react-app`
- Vite: `npm create vite@latest`

4. Execution Flow:

`index.html` → `main.jsx` → `App.jsx` → Components → UI

5. JSX Working:

JSX is converted into `React.createElement` using Babel.

6. State and Props:

- Props: Data passed from parent to child
- State: Local component data

7. Hooks:

- `useState`: Manage state
- `useEffect`: Side effects

8. Event Handling:

Use `onClick`, `onChange` etc. with functions

9. Conditional Rendering:

Use ternary operators or `&&`

10. Lists & Keys:

Use `map()` and unique keys

11. Forms:

Controlled components using state

12. API Calls:

Use `fetch` or `axios` inside `useEffect`

13. Routing:

Use `react-router-dom`

14. Best Practices:

- Use functional components
- Keep components small
- Use proper folder structure